

Toward Principled Threshold Selection for Blink Region Detection in Biological Signals: A Cross-Dataset Benchmark with LLM-Guided Candidate Derivation

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Abstract

Blink detection from electroencephalography and related biological signals remains important for two seemingly opposite reasons: blinks are a major physiological contaminant in EEG analysis, and they are also informative behavioral and neurophysiological markers linked to arousal, fatigue, cognitive load, and operational performance. Despite substantial progress in blink detection, the literature largely evaluates complete detectors rather than isolating the threshold-selection problem itself. This is a notable gap because many classical, hybrid, and real-time blink detectors depend critically on threshold choice, yet threshold rules are often dataset-specific, weakly justified, or folded into larger pipelines. We therefore propose a benchmark study focused on threshold selection for blink-region detection across heterogeneous datasets. The benchmark compares global mean-standard deviation thresholds, robust median absolute deviation thresholds, adaptive local thresholds, literature-derived hybrid thresholding, and an LLM-guided family of new derivative thresholds under a common preprocessing, matching, and evaluation protocol. The primary supervised setting uses naturalistic frontal-EEG blink annotations in a sustained-attention driving corpus and recommends extension to a multimodal ocular-activity dataset with synchronized EEG, eye tracking, and high-speed video; a motor-imagery EEG corpus is retained only as an auxiliary false-positive stress test because it is not natively blink-labeled. Performance is assessed by event-level precision, recall, and F1 using overlap-based matching, which is appropriate for sparse interval events with uncertain boundaries. The anticipated contribution is not merely a stronger detector, but a principled account of which threshold families remain stable under subject variability, operational drift, and domain shift.

1 Introduction

Electroencephalography is routinely affected by ocular activity, and blink contamination remains one of the oldest and most persistent obstacles in EEG analysis. At the same time, spontaneous and task-related blinks are themselves scientifically meaningful. Prior work links blink-derived measures to fatigue, drowsiness, attention, cognitive load, and operational state, and more recent blink-related EEG studies show that blink-linked neural activity can itself index cognitive demands in realistic tasks (Croft and Barry, 2000; Kleifges et al., 2017; Alyan et al., 2023). These dual roles make blink detection a foundational problem rather than a preprocessing detail: researchers need detectors that are accurate enough for artifact handling, but also reliable enough to preserve blink events as analyzable signals in their own right.

Thresholding remains central to this problem because even sophisticated systems frequently begin with amplitude, prominence, duration, or candidate-window gates. Threshold-based designs are attractive in practice because they are interpretable, fast, and deployable in few-channel and real-time environments. That is especially important for frontal wearable EEG and neuroergonomic

settings, where channel count, latency, and computational budget are constrained. However, thresholding is also where many detectors fail: a threshold learned from background-dominated samples may drift toward over-detection, while a threshold tuned to one subject or one experiment can collapse under inter-subject variability or under distributions that include eye movements, muscle activity, or epileptiform events (Chang et al., 2016; Kleifges et al., 2017; Agarwal and Sivakumar, 2019; Tran et al., 2021; M. Wang et al., 2022; Y. Zhang et al., 2023).

The central hypothesis of this paper is that threshold selection should be treated as a first-class experimental variable. Instead of asking only which detector wins, we ask which threshold family remains stable when preprocessing, channels, event matching, and datasets are held constant. This paper therefore frames blink detection as a threshold-selection benchmark over interval events. The benchmark explicitly separates signal conditioning, candidate peak extraction, threshold family, region formation, and event matching. It further adds a tightly bounded LLM-assisted ideation module in which the model suggests new threshold derivatives from the literature, but only human-vetted formulas are benchmarked (J. Cao et al., 2021; Saito and Rehmsmeier, 2015; Salles et al., 2024; Gottweis et al., 2025).

The contributions are fourfold. First, we formalize a threshold taxonomy for blink-region detection in biosignals. Second, we propose a cross-dataset protocol that distinguishes supervised blink benchmarking from out-of-domain robustness testing. Third, we justify overlap-based event scoring with precision, recall, and F1 as the main evaluation regime for sparse blink intervals. Fourth, we introduce an LLM-guided but human-constrained mechanism for proposing interpretable threshold derivatives that can be evaluated under the same benchmark as literature-derived baselines.

2 Related Work

The classical literature established the difficulty of ocular contamination and motivated automated statistical detection directly from EEG. Early work emphasized artifact removal and statistical screening rather than blink-region benchmarking per se, but it already showed that extreme-value behavior, channel context, and morphology matter (Croft and Barry, 2000; Klein and Skrandies, 2013). This historical backdrop explains why simple one-value global thresholds can be attractive, yet brittle.

A first important stream comprises single-channel or few-channel threshold-centered detectors. Chang et al. (2016) detect blink artifacts from a single prefrontal channel using digital filtering and rule-based decisions. Tran et al. (2021) present an explicitly thresholding-based EEG blink detector and compute a minimum amplitude threshold from transformed signal magnitude and signal standard deviation. Y. Zhang et al. (2023) move toward real-time frontal EEG detection with short windows and decision logic around potential blink boundaries. M. Wang et al. (2022) extend thresholding to harder clinical settings by optimizing multidimensional features in the presence of epileptiform discharges, and later work based on higher-order cumulants continues the move toward short-segment single-channel detection (G. Wang et al., 2025).

A second stream contains hybrid thresholding pipelines. BLINKER first identifies candidate intervals with a mean-based standard-deviation threshold and then rejects implausible candidates using robust amplitude-distribution criteria derived from the median absolute deviation (Kleifges et al., 2017). J. Cao et al. (2021) use cascaded thresholding over amplitude, amplitude displacement, and cross-channel correlation before applying a Gaussian mixture model. Agarwal and Sivakumar (2019) self-learn user-specific profiles in an unsupervised manner, while Valderrama et al. (2018) use iterative template matching and automatic threshold/template estimation for suppression in single-channel recordings. Guttman-Flury, Sheng, D. Zhang, et al. (2019) further emphasize adaptation

to inter- and intra-subject variability, underscoring that fixed thresholds are often not enough.

A third stream is the recent learning-based and multimodal literature. The recent review of deep learning in blink detection shows that the field is increasingly data-driven, yet also constrained by data imbalance, device heterogeneity, and generalization challenges (Xiong et al., 2025). For a threshold paper, this trend matters for two reasons. First, it confirms that threshold methods remain useful as transparent baselines. Second, it motivates a bridge strategy in which LLMs or other generative tools are not used to replace evaluation, but to propose candidate formulas that remain interpretable and falsifiable (Gottweis et al., 2025).

What is still missing is a benchmark that compares threshold families themselves under shared data and shared scoring. Prior papers generally optimize their own end-to-end pipeline and report strong performance in their own settings, but they do not isolate threshold choice as the controlled independent variable. That is the gap targeted here.

2.1 Problem Definition

Let a blink annotation be an interval $g_i = [s_i, e_i]$ and a detector output be an interval $d_j = [\hat{s}_j, \hat{e}_j]$. A detected region is counted as a true positive if it overlaps at least one unmatched ground-truth region. Unmatched detections are false positives; unmatched ground-truth regions are false negatives. This event-level treatment follows the logic of interval-based detection rather than pointwise accuracy, which is appropriate because blink boundaries are uncertain, annotation granularity varies, and non-event samples dominate the timeline (Saito and Rehmsmeier, 2015; Salles et al., 2024).

2.2 Datasets

The primary naturalistic dataset is the sustained-attention driving corpus of Z. Cao et al. (2019), which contains 62 sessions of 32-channel EEG recorded during a 90-minute lane-keeping task. Its official event schema includes deviation onset, response onset, and response offset, and the published preprocessed version explicitly notes manual removal of apparent eye blinks. Therefore, this corpus is suitable for the present study only in its raw form and only when paired with externally available blink-region annotations.

The second dataset named in the project brief requires correction. PMID 30325349 refers to the Scientific Data motor-imagery corpus of Kaya et al. (2018), not to a blink-annotated benchmark. It is therefore inappropriate as a primary supervised blink dataset. In the present design it is repurposed as an auxiliary robustness corpus for false-positive stress testing: because it contains frontal EEG but lacks blink-event labels, it can help measure how often a threshold family spuriously fires in out-of-domain EEG.

Because the paper aims to be scientifically robust rather than merely brief-compliant, we recommend adding the 2025 multimodal ocular-activity dataset by Guttmann-Flury, Sheng, and Zhu (2025) as a companion benchmark. Its synchronized EEG, eye tracking, and high-speed video make it the strongest official source for validating blink-region decisions and studying boundary uncertainty.

2.3 Preprocessing

All threshold families are evaluated under the same preprocessing backbone. We recommend using frontal or prefrontal channels only, after polarity normalization if needed, detrending, and a blink-oriented band-pass in the low-frequency region. Candidate maxima are then detected from the preprocessed signal or a derived blink component, and candidate windows are expanded by local valley search or hysteresis rules. This standardization is necessary because the literature shows that

filter choice, candidate signal choice, and reference scheme can strongly alter the apparent success of a threshold (Chang et al., 2016; Kleifges et al., 2017; Tran et al., 2021; Y. Zhang et al., 2023).

2.4 Threshold Families

We propose five threshold families. The first is a classical mean-standard-deviation family, $\theta = \mu + k\sigma$, included because BLINKER’s initial candidate gate effectively uses a variant of that logic after filtering. The second is a mean-plus-robust-spread family, $\theta = \mu + k \cdot 1.4826 \text{ MAD}$, included because it matches the simplified code variants discussed in the project brief. The third is a median-plus-robust-spread family, $\theta = \tilde{x} + k \cdot 1.4826 \text{ MAD}$, which is expected to be less sensitive to skewed baselines and large outliers. The fourth is an adaptive rolling-median/rolling-MAD hysteresis family, with a high onset threshold and a lower offset threshold, proposed as the main new derivative because it preserves local adaptability while retaining interpretability. The fifth is a literature-derived hybrid family, represented by cascaded thresholding prior to GMM or template screening, as in J. Cao et al. (2021) and related adaptive systems.

The key critique motivating families two through four is that the background-dominated baseline of continuous EEG can make a single global threshold too permissive. If the threshold is estimated mostly from non-blink samples, then drift, small ocular movements, muscle transients, and transient nonstationarities can cross the gate. The benchmark therefore asks whether robustness should be introduced through the location statistic, the scale statistic, the time scale, or the decision logic.

3 Methods

3.1 LLM-Guided Candidate Derivation

The LLM component is deliberately narrow. The model is not asked to label data or choose winners. Instead, it receives a structured literature summary and proposes mathematically explicit threshold variants, for example asymmetry-aware onset/offset rules, quantile-MAD hybrids, or drift-compensated local thresholds. Human researchers then filter these suggestions for interpretability, computational feasibility, and absence of label leakage before implementation. This design aligns with recent peer-reviewed work on scientific hypothesis generation, which treats LLMs as idea generators whose outputs must still be experimentally vetted (Gottweis et al., 2025).

3.2 Evaluation

The primary metrics are event-level precision, recall, and F1 under non-zero-overlap matching, with secondary analyses using temporally tolerant variants or IoU-weighted summaries. Precision, recall, and F1 are preferred because blink events are sparse relative to non-blink samples; raw accuracy would be inflated by true negatives and would obscure detector behavior on the rare events of interest. Soft temporal metrics are reported secondarily because interval boundaries are often approximate and neighboring detections can still be operationally useful (Saito and Rehmsmeier, 2015; Salles et al., 2024).

Per-session estimates should be macro-averaged, with bootstrap confidence intervals across sessions and a paired nonparametric comparison of threshold families on session-level F1. For the motor-imagery corpus, the relevant endpoint is not F1 but false-positive burden per minute and per hour, since the corpus is used only for robustness stress testing.

Table 1: Planned result table. Numerical cells are placeholders for benchmark output.

Threshold family	Driving P/R/F1	Ext. F1	Kaya FP/h	Anticipated failure mode
Mean + SD	[fill]/[fill]/[fill]	[fill]	[fill]	Baseline drift and small eye movements
Mean + robust MAD	[fill]/[fill]/[fill]	[fill]	[fill]	Permissive if mean remains biased
Median + robust MAD	[fill]/[fill]/[fill]	[fill]	[fill]	Missed low-amplitude blinks
Rolling median/MAD + hysteresis	[fill]/[fill]/[fill]	[fill]	[fill]	Boundary fragmentation if window is poorly chosen
Cascaded threshold + GMM/template screening	[fill]/[fill]/[fill]	[fill]	[fill]	Complexity and tuning overhead

4 Experiments and Results

Because no new benchmark was executed in this conversion step, the prose below is submission-ready but the numerical cells are intentionally left as placeholders for post-run insertion.

We evaluate the five threshold families under identical preprocessing, candidate generation, and event matching. Hyperparameters are tuned only on the training fold of each dataset. The primary supervised benchmark uses session-wise cross-validation on the driving dataset with manual blink-region annotations; the multimodal ocular dataset is used for external validation; and the Kaya motor-imagery dataset is used only to assess out-of-domain false positives (Z. Cao et al., 2019; Kaya et al., 2018; Guttmann-Flury, Sheng, and Zhu, 2025).

Based on the reviewed literature, we expect global mean-based thresholds to be competitive in recall but unstable in precision, especially under naturalistic drift and mixed ocular activity. We expect median-plus-MAD and rolling-MAD hysteresis to improve precision without catastrophic recall loss, because these families more directly address skew, outliers, and local baseline drift. We further expect hybrid systems to remain strongest when data include structured confounds such as eye movements, muscle activity, or epileptiform patterns, but at the cost of greater pipeline complexity.

Ablation analysis should then ask three specific questions. First, does replacing the mean with the median improve cross-subject stability? Second, does making the scale estimator robust improve precision more than recall degrades? Third, does local adaptation outperform global calibration when evaluated on long sessions with nonstationary vigilance and drift? The answer to those questions matters more scientifically than the headline win of any single detector, because it reveals what kind of thresholding assumption is actually justified for blink-region detection.

5 Discussion

The most important methodological point is that thresholds are not just tuning knobs; they encode assumptions about the signal distribution. A global mean-plus-spread threshold assumes a stable background process and implicitly treats blink amplitude as a large deviation from that background. Yet the blink literature repeatedly shows that subject variability, state variability, and mixed non-blink ocular events violate that assumption. This is why robust statistics, adaptive windows, templating, morphology rules, and hybrid screening keep reappearing across otherwise different papers.

The second point concerns the software baseline question. If a peak-detector baseline is desired for completeness, a citable academic baseline should be favored over an unattributed code lineage.

In practical terms, that means replacing a software-only MNE/Yoder provenance claim with a peer-reviewed peak-detection comparator such as AMPD (Scholkmann et al., 2012), or else clearly labeling the MNE routine as a software baseline outside the formal manuscript references. Doing so preserves the peer-reviewed reference requirement without losing the practical comparison that motivated the question.

The third point concerns the role of LLMs. In this setting, the strongest use of an LLM is not to decide which blink candidate is correct, but to expand the design space of interpretable threshold formulas. That is scientifically defensible because the generated rules remain human-auditable and experimentally falsifiable. In other words, the LLM is a hypothesis generator, not a substitute evaluator.

The main limitation of the drafted study is that its strongest driving-data result depends on the existence and quality of manual blink-region annotations, since the official driving dataset does not ship native blink-event labels. A second limitation is that the Kaya motor-imagery corpus, while useful for stress testing, cannot answer supervised blink-detection questions by itself. These constraints are precisely why the multimodal ocular dataset is recommended as an official companion benchmark.

6 Conclusion

This draft argues that blink detection in biological signals should be studied not only as an end-to-end detector problem, but also as a threshold-selection problem. The literature supports the importance of blink analysis for artifact handling, cognitive-state inference, and real-time neuroergonomics, but it also shows that thresholding assumptions are frequently hidden inside larger pipelines. A benchmark that isolates threshold families, evaluates them under shared overlap-based event scoring, and supplements literature-derived baselines with LLM-generated but human-vetted candidates can make the field more cumulative, more interpretable, and easier to reproduce.

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